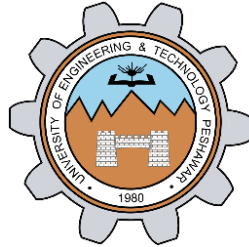


Interfacing LEDs to 8051 and displaying Binary Patterns

LAB # 02



Spring 2024

CSE-307L Microprocessor Based System Design Lab

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Registration No.: **21PWCSE2059**

Class Section: **C**

“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

Dr. Asif Ali Khan

Date:

25th February 2024

Department of Computer Systems Engineering
University of Engineering and Technology, Peshawar

Task 1:

D:\Uni\MBSD - Lab\Keil_Work\Lab2_Project\task1.uvproj - uVision [Non-Commercial Use License]

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x1e
r6	0x00
r7	0x00
sys	
a	0x1e
b	0x00
sp	0x07
sp_max	0x07
dptr	0x0000
PC \$	C:0x0009
states	33
sec	0.00001650
psw	0x00

Disassembly

```
2:      MOV A, #0
C:0x0000 7400  MOV     A, #0x00
3:      MOV R2, #10
4:
5:  AGAIN:
C:0x0002 7A0A  MOV     R2, #0x0A
6:      ADD A, #03
C:0x0004 2403  ADD     A, #0x03
7:      DJNZ R2, AGAIN
```

task1.a51

```
1
2  MOV A, #0
3  MOV R2, #10
4
5  AGAIN:
6  ADD A, #03
7  DJNZ R2, AGAIN
8  MOV R5, A
9
10 END
```

Task 2:

D:\Uni\MBSD - Lab\Keil_Work\Lab2_Project\task2.uvproj - uVision [Non-Commercial Use License]

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
sys	
a	0x55
b	0x00
sp	0x07
sp_max	0x07
dptr	0x0000
PC \$	C:0x000B
states	2132
sec	0.00106600
psw	0x00

Disassembly

```
10:      DJNZ R3, NEXT
C:0x0009 DBF9  DJNZ   R3, NEXT (C:0004)
C:0x000B 00  NOP
C:0x000C 00  NOP
```

task2.a51

```
3
4  NEXT:
5  MOV R2, #70
6
7  AGAIN:
8  CPL A
9  DJNZ R2, AGAIN
10 DJNZ R3, NEXT
11
12 END
```

Task 3:

Code:

Using Assembly

```
labtask.c lab1.a51
1 ORG 00H
2     HERE:
3     MOV P0, #01H
4     ACALL DELAY
5     MOV P0, #02H
6     ACALL DELAY
7     MOV P0, #04H
8     ACALL DELAY
9     MOV P0, #08H
10    ACALL DELAY
11    MOV P0, #10H
12    ACALL DELAY
13    MOV P0, #20H
14    ACALL DELAY
15    MOV P0, #40H
16    ACALL DELAY
17    MOV P0, #80H
18
19    MOV P0, #40H
20    ACALL DELAY
21    MOV P0, #20H
22    ACALL DELAY
23    MOV P0, #10H
24    ACALL DELAY
25    MOV P0, #08H
26    ACALL DELAY
27    MOV P0, #04H
28    ACALL DELAY
29    MOV P0, #02H
30
```

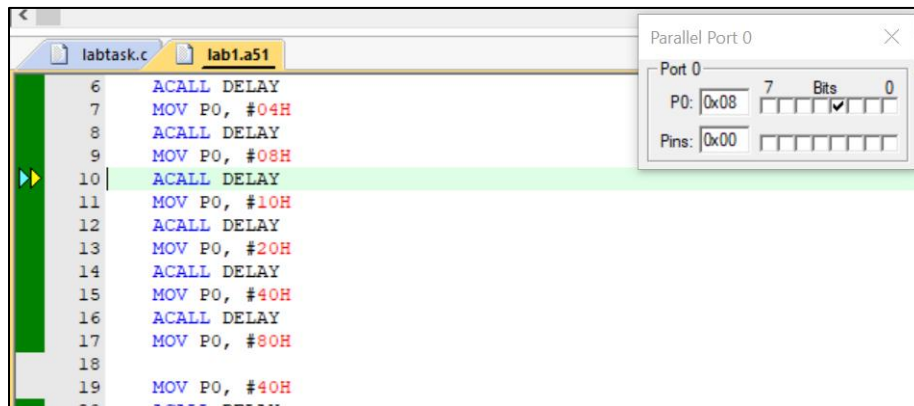
```
28    ACALL DELAY
29    MOV P0, #02H
30
31    SJMP HERE
32
33    DELAY:
34        MOV R0, #255
35        MOV R1, #255
36
37        LOOP1:
38            LOOP2:
39                DJNZ R0, LOOP2
40            DJNZ R1, LOOP1
41
42    RET
```

Using C

```
task3.c
1  #include<reg51.h>
2
3  int a, i, j;
4
5  void delay(){
6
7      for(i = 0; i<255; i++){
8          for(j = 0; j<255; j++){
9              ;
10             }
11         }
12
13     void main(){
14
15         while(1){
16             P0 = 0x01;
17             delay();
18
19             P0 = 0x02;
20             delay();
21
22             P0 = 0x04;
23             delay();
24
25             P0 = 0x08;
26             delay();
27
28             P0 = 0x10;
29             delay();
```

```
29
30         P0 = 0x20;
31         delay();
32
33         P0 = 0x40;
34         delay();
35
36         P0 = 0x80;
37         delay();
38
39         P0 = 0x40;
40         delay();
41
42         P0 = 0x20;
43         delay();
44
45         P0 = 0x10;
46         delay();
47
48         P0 = 0x08;
49         delay();
50
51         P0 = 0x04;
52         delay();
53
54         P0 = 0x02;
55         delay();
56     }
57 }
```

Output:



Proteus Simulation:

